MILITARY SPECIFICATION

COATING, ELASTOMERIC, POLYURETHANE, RAIN EROSION RESISTANT, FOR EXTERIOR AIRCRAFT USE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for two types of rain erosion resistant coatings for glass fabric base plastic laminates and other aircraft parts.

1.2 Classification. The coating shall be furnished in the following types and colors, as specified (see 6.2):

Type I - Two component rain erosion resistant coating to be furnished in two separate packages.

Type II - Two component antistatic, rain erosion resistant coating to be applied over type I for antistatic purposes and be furnished in two separate packages.

1.2.1 Colors. The color of the polyurethane coating shall be as listed below:

Type I		Type II	
FED-STD-595	Color number	FED-STD-595	Color number
17038	black	17038	black
16357	gray		
17925	white		

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8010

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SPECIFICATIONS

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Federal

L-P-383	-	Plastic Material, Polyester Resin, Glass Fiber Base, Low Pressure Laminated.
QQ-A-250/5	-	Aluminum Alloy, Plate and Sheet, General Specification for.
PPP-P-1892	-	Paint, Varnish, Lacquer and Related Materials; General Specification for Packaging, Packing and Marking

Military

MIL-P-5425	-	Plastic; Acrylic, Heat Resistant.
MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance.
MIL-T-5624	-	Turbine Fuel, Aviation Grades JP-4 and JP-5.
MIL-R-7705	-	Radome, General Specification for.
MIL-C-8514	-	Coating Compound, Metal Pretreatment, Resin-Acid.
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-P-23377	-	Primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant, Type I.
MIL-L-23699	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-P-25421	-	Plastic Materials, Glass Fiber Base-Epoxy Resin, Low Pressure Laminated.
MIL-T-81772	-	Thinner, Aircraft Coating.
MIL-H-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft.

STANDARDS

Federal

FED-STD-141	-	Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.
FED-STD-313	-	Material Safety Data Sheets; Preparation and Submission of.
FED-STD-595	-	Colors.

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein.

PUBLICATIONS

BUREAU OF MEDICINE AND SURGERY

BUMEDINST 6260.16A - Polyurethane Paints and Other Substances Containing Isocyanates; Measures of Control of Health Hazards Related to.

(Application for copies should be addressed to Commander, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 882	-	Tensile Properties of Thin Plastic Sheeting.
ASTM D 1200	-	Viscosity of Paints, Varnishes and Lacquers by Ford Viscosity Cup.
ASTM D 1210	-	Fineness of Dispersion of Pigment-Vehicle Systems
ASTM D 1640	-	Drying, Curing or Film Formation of Organic Coatings at Room Temperature
ASTM D 2615	-	Free Toluene Diisocyanate in Urethane Polymers by Distillation.
ASTM G 26	-	Operating Light and Water-Exposure Apparatus (Xenon-arc type) for Exposure of Nonmetallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Qualification. The coatings furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.3 and 6.3). Any change in the formulation of a qualified product will necessitate its requalification. The material supplied under contract shall be identical, within manufacturing tolerances, to the product receiving qualification.

3.2 Materials. The ingredients used in the manufacture of these products shall conform to the applicable requirements as specified herein.

3.2.1 Appearance. The rain erosion resistant and antistatic materials (in the container) shall be uniform, homogeneous and free of skins, lumps, gels or coarse particles. There shall be no separation of ingredients which cannot be dispersed readily. There shall be no gassing or swelling of the containers.

3.2.2 Working properties. Type I and II coatings shall be suitable for brush or spray application to plastic laminates, composites and aluminum alloy substrates meeting Military specifications. The material shall be mixed in accordance with the manufacturer's instructions.

3.3 Components. Unless otherwise specifically approved by the acquiring activity (see 6.2), the Type I coating shall consist of a rain erosion resistant Component A with curing agent Component B, and the Type II coating shall consist of antistatic Component A and curing agent Component B. Curing agent Component B is not antistatic until it is mixed with antistatic Component A.

3.4 Composition.

3.4.1 Type I coating. The rain erosion Type I coating shall be a two component system consisting of component A which is a polyurethane elastomeric base component, and component B, which shall be a ketimine type curing agent (blocked diamine). They shall be mixed in a ratio of 2 volumes of Component A to 1 volume of Component B.

3.4.2 Type II coating. The antistatic Type II coating shall be a two component system consisting of antistatic component A which is a polyurethane elastomeric base component which meets the surface resistivity requirements of 3.7.16 and a component B which shall be a ketimine type curing agent (blocked diamine).

3.4.3 Components A and B.

3.4.3.1 Component A. The vehicle in component A for both Type I and II coatings shall be polyether type polyurethane prepolymers that exhibit the characteristic infrared absorption curve of Figure 1 and meet the requirements of Table I. The pigmented portion shall meet the requirements in Table I when tested as specified in Table VII (see 4.8.1). Component A for Type II shall meet the antistatic requirements of 3.7.16.

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3.4.3.2 Component B curing agent. The curing agent, a ketimine type blocked diamine, shall be a clear, amber colored solution that exhibits the characteristic infrared absorption curve of Figure 2. Component B curing agent is used with Component A of Type I and antistatic Component A of Type II (see 4.8.1).

3.5 Viscosity (Ford cup). The initial viscosity of the admixed coatings shall be a minimum of 30 seconds using a No. 4 Ford cup (77°F). When Part A and Part B are mixed, the viscosity begins to increase (see 4.8.2).

3.6 Toxicity. The coating components shall have no adverse effect on the health of personnel when used in accordance with 5.2 for its intended purpose. Questions pertinent to this effect shall be referred by the acquiring activity to the appropriate medical service who will act as an advisor to the acquiring activity. The manufacturer shall certify that the coating contains no substance known to be toxic to the user under normal conditions of use. Material safety data sheets shall be prepared and submitted in accordance with FED-STD-313, one copy of which shall be forwarded to the qualifying activity.

3.7 Physical properties.

3.7.1 Storage stability. The materials shall be capable of meeting the requirements specified herein, when stored separately in their original unopened containers for a minimum of one year at a maximum temperature of 90°F after date of manufacture.

3.7.2 Accelerated storage stability. The pigmented components shall be subjected to accelerated storage in accordance with 4.8.3. At the end of the storage period there shall be no gassing that causes swelling of the cans, the component materials shall be free of lumps, skins and gels, and when stirred shall disperse readily to a smooth homogeneous mixture.

TABLE I

Component A Requirements.

	Black	White	Gray
Pigment (Based on Non-volatile)	5% max.	16% max.	14% max.
Binder (Based on Non-volatile)	95% max.	84% max.	86% max.
Fineness of Grind, PC (Mixed Topcoat)	6 min.	6 min.	6 min.
Free Toluene Diisocynate (Pigment Component)	0.4% max.	0.4% max.	0.4% max.
Color Number (FED-STD-595)	17038	17925	16357

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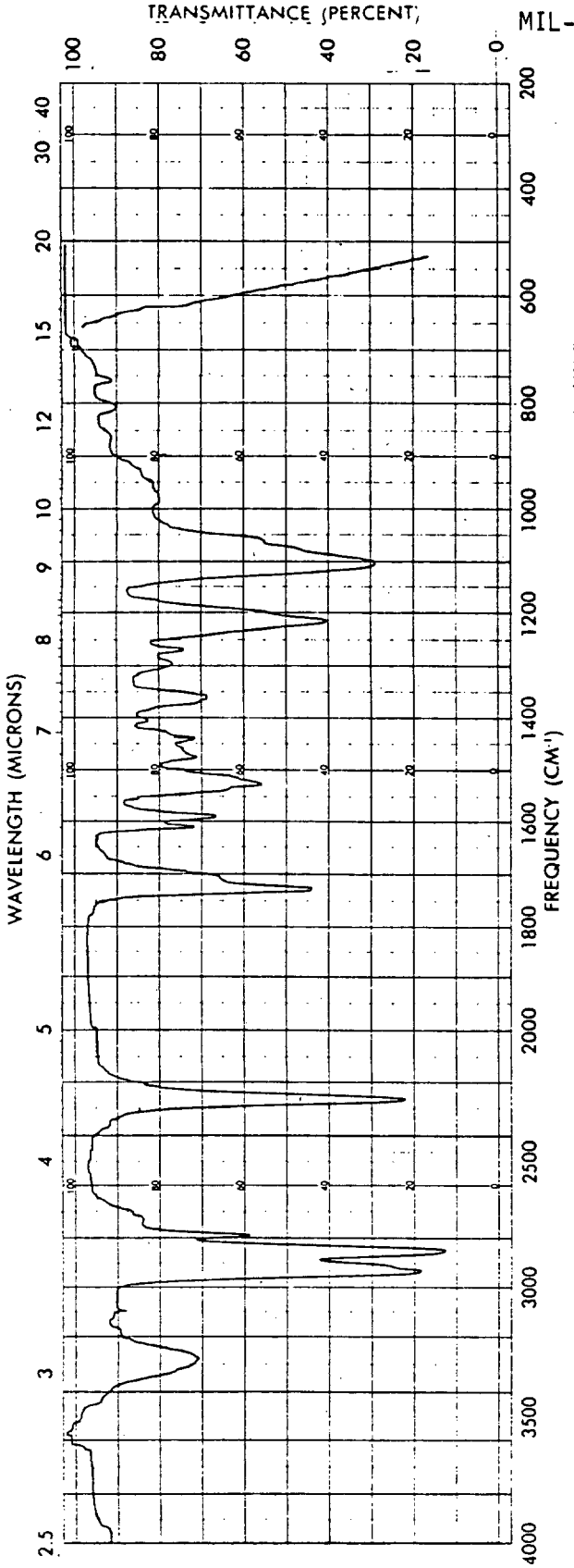


FIGURE 1. Infrared Absorption Spectra of Base Component (Polyurethane Prepolymer).

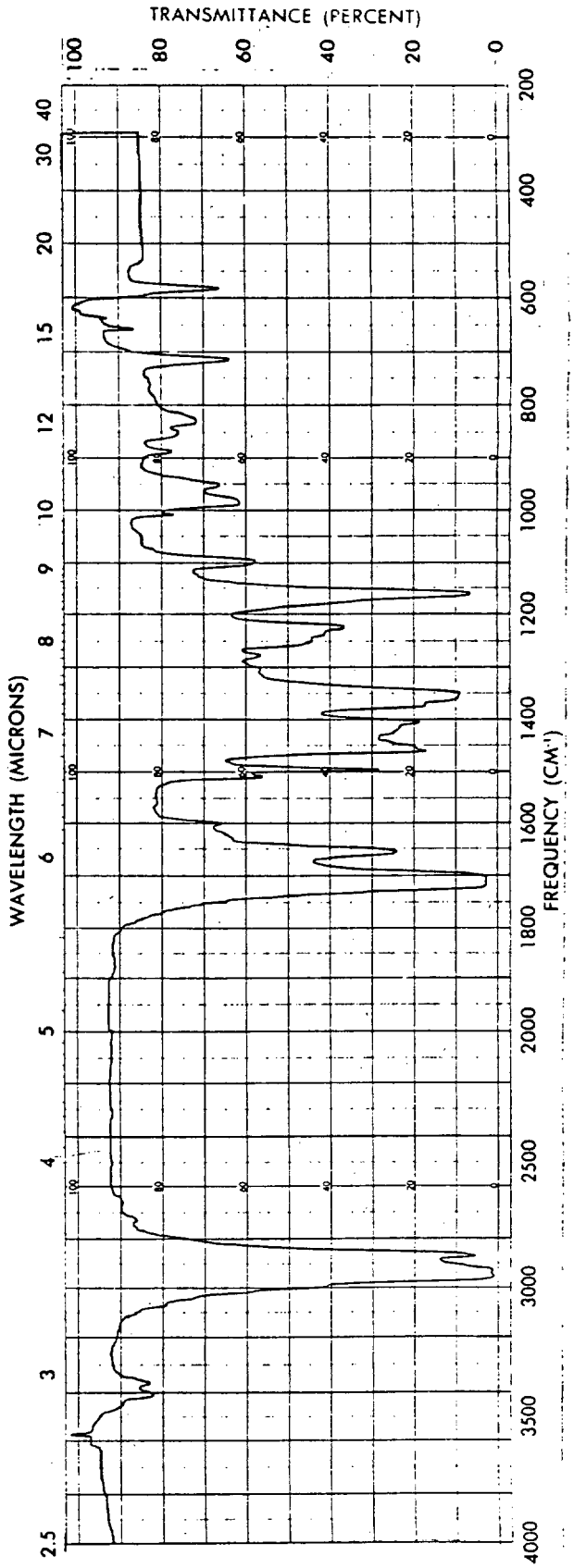


FIGURE 2. Infrared Absorption Spectra of Curing Agent (Blocked Diamine).

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3.7.3 Pot life. The pot life of the mixed rain erosion coating, when tested in accordance with 4.8.12, shall be at least 2 hours at 73.5°F + 2°F (23°C + 1.1°C) and 50% + 10% relative humidity when maintained in a full closed container. The working properties of the mixture and surface appearance of a panel sprayed with the material shall meet all requirements of Section 3, except for storage stability (3.7.1) and accelerated storage stability (3.7.2) requirements.

3.7.4 Free film strength. Free film strength of the applied rain erosion resistant and antistatic coating, prepared and tested as specified in paragraph 4.8.5, shall meet the requirements as listed in table II.

TABLE II.

Free film strength.

		168 hr. Cure	3 hrs 220°F
Tensile Strength	p.s.i. min	450	1600
Elongation-at-break	% min.	450	500
300% modulus	p.s.i. min.	-	650

3.7.5 Drying time. The film, after application to a panel, shall dry hard within a maximum time of eight hours (see 4.6).

3.7.6 Dry-to-handle. Twenty-four hours after application the coating shall be capable of being handled without any abnormal care and shall possess sufficient film strength to maintain complete film integrity while being handled. A slight tack which can be removed by a light dusting with talc is permissible.

3.7.7 Peel adhesion. The peel adhesion shall be a minimum of six pounds per inch of width. Film breakage at lower values than six pounds without the film peeling from the panel shall be considered passing (see 4.8.4).

3.7.8 Applied film properties. The polyurethane film shall be free from pin-holes, cracks, bubbles or other film irregularities. A slight after tack shall be permissible (see 4.8.13).

3.7.9 Low temperature flexibility. The coating shall show no cracking or loss of adhesion when tested in accordance with 4.8.6.

3.7.10 Water resistance. The coating after immersion in water shall show no evidence of blistering, checking or visible color change (see 4.8.7).

3.7.11 Fuel resistance. The coating after immersion in JP-4 and JP-5 fuels conforming to MIL-T-5624 shall show no evidence of blistering, checking, visible color change or other defects.

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3.7.12 Fluid resistance. The polyurethane film shall withstand immersion for 4 hours in the following fluids at room temperature: MIL-L-23699 lubricating oil, MIL-H-5606 Hydraulic fluid, and MIL-H-83282 hydraulic fluid. Four hours after removal, the film shall not exhibit any blistering, softening, dark staining, or other defects (see 4.8.15).

3.7.13 Accelerated weather resistance. The coatings, when exposed to accelerated weathering for 500 hours, shall show no cracking, checking, embrittlement or loss of adhesion.

3.7.14 Electrical transmission. The coating shall meet the requirements as specified in Table III (see 4.8.9).

TABLE III.

Electrical Transmission

	After 7 day Cure at 50 ± 10% RH and 75 ± 5°F	After 7 day Cure plus 7 days at 90% RH and 100°F
Electrical Transmission (% min.)	90	85

3.7.15 Hydrolytic stability. The film shall show no softening or loss of adhesion, when tested as specified in 4.8.10.

3.7.16 Surface resistivity. The surface resistivity of the Type II coatings shall not be less than 0.5 nor more than 15 megohms per square when tested as specified in 4.8.11.

3.7.17 Rain erosion resistance. Types I and II coatings, when tested as specified in 4.8.17, shall meet the minimum time erosion requirements for the corresponding coating thickness specified in table IV.

3.7.18 Abrasion resistance. The Types I and II coatings applied to test panels and cured as specified, when tested as specified in 4.6.18, shall meet the minimum abrasion resistance requirements in Table V.

3.7.19 Strippability. Newly applied, fully cured and weathered Types I and II coatings shall be capable of being stripped when processed as specified in 4.8.16.

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TABLE IV. Rain Erosion Resistance - Whirling Arm.

Test conditions: 500 MPH, 1/2 INCH/HOUR SIMULATED RAINFALL 1/
(1.8mm dia. droplets)

COLOR	AVERAGE COATING THICKNESS MILS (CM)	MINIMUM TIME OF EXPOSURE (MINUTES)
White	12(.033)	150
	14(.035)	180
Gray	12(.033)	150
	14(.035)	180
Black	12(.033)	130
	14(.035)	150

1/ TEST BASED ON AN AVERAGE OF 4 GLASS-EPOXY SUBSTRATES.
SOME SUBSTRATE CRUSHING IS ACCEPTABLE WHEN TESTING SPECIMENS.

TABLE V. Abrasion Resistance - S.S. White Test. 2/

COLOR	FILM THICKNESS MILS (CM)	MINIMUM PENETRATION TIME 1/ (SEC.)	MINIMUM ABRASION EFFICIENCY (SEC./MIL.)	MINIMUM ABRASIVE INDEX (Y)
WHITE	13(.034)	1600	123	2.68
GRAY	13(.034)	1500	115	2.55
BLACK	13(.034)	1400	108	2.38

1/ AVG. OF FIVE PENETRATION TESTS

2/ S.S WHITE (MODEL K) ABRADER (OR EQUAL)
ABRASIVE NO. 10 - POWDER FLOW AT 6-60 PSI (413.7 kPa)

$$\text{ABRASIVE INDEX VALUE} \quad Y = \frac{t_1}{s_1} \times \frac{s_2}{t_2}$$

t_1 = Time in seconds to erode film under test
t_2 = Time in seconds to erode control film
s_1 = Thickness of film under test in mils (X .0025=cm)
s_2 = thickness of control film in mils (X .0025=cm)

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 Qualification inspection. Qualification inspection shall be as specified in Table VI.

TABLE VI. Qualification Inspection.

Inspection	Requirement paragraph	Test method paragraph
Free Toluene Diisocyanate	3.4.3.1	4.6
Vehicle solids	3.4.3.1	4.6
Fineness of grind	3.4.3.1	4.6
Color	3.4.3.1	4.6
Viscosity, Ford Cup, Coating at 25°C ± 0.5°C	3.5	4.8.2
Storage stability	3.7.1	4.6
Accelerated storage stability	3.7.2	4.8.3
Pot life	3.7.3	4.8.12
Free film strength	3.7.4	4.8.5
Drying time	3.7.5	4.6
Peel adhesion	3.7.7	4.8.4
Applied film properties	3.7.8	4.8.13
Low temperature flexibility	3.7.9	4.8.6
Water resistance	3.7.10	4.8.7
Fuel resistance	3.7.11	4.8.14
Fluid resistance	3.7.12	4.8.15
Accelerated weather resistance	3.7.13	4.8.8
Electrical transmission	3.7.14	4.8.9
Hydrolytic stability	3.7.15	4.8.10
Surface resistivity	3.7.16	4.8.11
Rain erosion resistance	3.7.17	4.8.17
Abrasion resistance	3.7.18	4.8.18
Strippability	3.7.19	4.8.16

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4.3.1 Qualification samples. The test samples shall consist of two gallons of Component A and one gallon of component B of the coating material in the white, gray and black colors (black only for Type II). The material shall be furnished in containers of the type to be used in filling contract orders. Samples shall be identified as follows and forwarded to the Naval Air Development Center, Aircraft and Crew Systems Technology Directorate, Code 60622, Warminster, PA 18974.

- Qualification test samples
- Specification MIL-C-85322 (Type I, White, Gray or Black (as applicable), (Type II, Black)
- Coating, Elastomers, Polyurethane, Rain Erosion Resistant
- Manufacturer's name and product number
- Submitted by (name and date) for qualification testing

4.3.2 Test report. In addition to the qualification test samples, the manufacturer shall furnish a test report showing that the material satisfactorily conforms to the requirements of this specification. Material safety data sheets shall be prepared and submitted in accordance with FED-STD-313.

4.3.3 Retention of qualification. In order to retain qualification of a product approved for listing on the Qualified Products List (QPL), the manufacturer shall verify by certification to the qualifying activity that the manufacturer's product complies with the requirements of this specification. The time of periodic verification by certification shall be in two-year intervals from the date of original qualification. The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specification requirements.

4.4 Quality conformance inspection. The quality conformance inspection shall consist of all the requirements specified in Section 3, with the exception of storage stability (3.7.1), rain erosion resistance (3.7.17), electrical transmission (3.7.14), and surface resistivity (3.7.16).

4.4.1 Sampling. At least one quart of each component of the coating material shall be selected at random from each batch by an authorized government representative and forwarded to the Naval Air Development Center, Attn: Code 60622, Warminster, PA 18974 for retention and possible confirmation testing. A batch shall consist of all coating material manufactured during one continuous operation and forming part of one contract or order for delivery.

4.4.1.1 Sampling for tests. Samples for tests shall consist of one complete unopened kit (see 5.1.1.1) selected at random from each lot. Containers shall only be opened when being tested.

4.4.1.2 Sampling for visual inspection of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105 and subjected to the examinations specified in 4.4.1.4.

4.4.1.3 Lot formation. A lot shall consist of all polyurethane coating of the same color, manufactured at one time from one batch, forming part of one contract, and submitted for acceptance at one time.

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4.4.1.4 Examinations.

4.4.1.4.1 Examination of preparation for delivery. The samples selected in accordance with 4.4.1.1 shall be examined for proper filling or weight, markings, packaging and packing. Any package having one or more defects or under required fill shall be rejected. If the number of defective packages exceeds the Acceptable Quality Level of 2.5 percent defective in accordance with Inspection Level S-4 of MIL-STD-105, the lot represented by the sample shall be rejected.

4.4.2 Report of tests. The manufacturer shall submit a certified report to the Government representative for each batch showing the results including numerical values when applicable to all tests specified in Table I. Reports of numerical tests as satisfactory are not acceptable. Each separate component shall be identified with the name of its manufacturer and the manufacturer's trade name and formula number.

4.4.2.1 Nonvolatile and volatile composition. In lieu of reporting analytical results on the breakdown of the nonvolatile and volatile composition of the components, the manufacturer may report such results as "calculated" under the condition that he has carefully described by separate report, attached to manufacturer's test reports, the character and details of his production methods which in his opinion, guarantee that any suitable analysis made by the Government will yield acceptable results.

4.4.2.2 Prepolymer and curing agent. The manufacturer shall submit a notarized certification signed by a responsible official of its management, attesting that the prepolymer and curing agent of the coating being furnished, regardless of the color, has met the storage stability test (3.7.1) when formulated. The specific test data on the prepolymer and curing agent combination may be obtained from the prime resin manufacturer for inclusion in the certification.

4.5 Additional testing. The Government reserves the right to conduct tests for storage stability and weather resistance or to conduct any or all tests of this specification at any time within 1 year from the date of manufacture of the coating as attested by the date appearing on the container's label. Samples for test shall be taken from previously unopened containers.

4.6 Test methods. The tests of this specification shall be conducted in accordance with Table VII and paragraphs 4.8.1 through 4.8.18 and the panels used prepared as specified in 4.7. Ingredient material submitted shall be tested to determine compliance with the applicable specification. Test conditions shall be as specified in the applicable test method or paragraph.

4.6.1 Standard conditions. Unless otherwise specified herein, all tests shall be conducted at a temperature of 75 ±5°F, and a relative humidity of 50 ±10 percent.

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TABLE VII. Test methods.

Requirement Paragraph No.	Test	FTMS 141 Method No.	ASTM Method
3.4.3.1	Free Toluene Diisocyanate of Prepolymer		D2615
3.4.3.1	Vehicle solids	4051	
3.4.3.1	Fineness of Grind		D1210
3.4.3.1	Color	4250	
3.5	Viscosity, Ford Cup, Coating at 25°C±0.5°C		D1200
3.7.1	Storage stability	3022	
3.7.4	Free Film Strength		D 882
3.7.5	Drying time		D 1640
3.7.9	Low temperature flexibility	6221	
3.7.13	Accelerated Weathering		G 26

4.7 Test specimen preparation. Unless otherwise specified herein, the test specimens shall be prepared as follows:

4.7.1 Test panels. Test panels for all tests except peel adhesion, electrical transmission, rain erosion, and abrasion resistance shall be aluminum clad aluminum alloy conforming to QQ-A-250/5, anodized in accordance with MIL-A-8625, Type I. Panels shall be 3 by 6 by 0.020 inches in size. The primer for glass fabric base plastic laminates shall be MIL-C-8514 or a two part polyvinyl butyral equal to or exceeding the requirements of MIL-C-8514. The primer for aluminum substrates shall conform to MIL-P-23377, Type I.

4.7.2 Coating of panels. The film thickness of the primer shall be 0.6 to 0.9 mils. The primer shall dry 60 minutes before application of the topcoat. The film thickness of the Type I topcoat shall be 12 to 14 dry mils. The admixed topcoat shall be thinned with MIL-T-81772 polyurethane thinner to a spray viscosity of 16 to 22 seconds through a No. 4 Ford cup. The topcoat shall be applied as 6 cross-coats allowing 5 minutes drying time between the two coats. Cure time shall be 168 hours before testing of Type I. Where bake cures are specified, the coatings shall air cure four days, and bake for 3 hours at 220°F, before testing of Type I.

4.7.2.1 Antistatic coating of panel. Apply 24 hours after applying Type I topcoat to 2 mils minimum thickness for antistatic testing.

4.8 Test methods. Test methods shall be as specified in Table V and 4.8.1 through 4.8.18.

4.8.1 Infrared absorption spectrum. Infrared absorption spectrum curves shall be obtained in a Perkin Elmer 521 Spectrophotometer of equal by following the manufacturer's instructions. A sealed cell and no oven should be used. The curves obtained shall meet the requirements of paragraph 3.4.3.

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4.8.2 Viscosity (mixed coatings). The initial viscosity of the mixed coatings shall be tested in accordance with ASTM D 1200 (see 3.5).

4.8.3 Accelerated storage stability. Unopened containers of the pigmented components shall be subjected to a minimum of 4 days of accelerated storage at 120 ±5°F. At the end of the 4 day storage period, the cans shall be visually inspected for swelling and gassing. The cans shall be opened and inspected for lumps, skins and gels; and the ease of the dispersing by paddle to a smooth homogeneous mixture shall be observed (see 3.7.2).

4.8.4 Peel adhesion.

4.8.4.1 Peel adhesion test panels. Test panels for the peel adhesion test shall be flat, glass-fiber base laminates conforming to any type and fabric number of MIL-P-25421. The glass laminate panels shall be of low void content, dense, and representative of high quality plastic laminated structures. They shall be lightly sanded with 180-240 grit abrasive paper to remove the glossy finish and wiped clean with a clean cloth saturated in a solvent such as toluene or methyl ethyl ketone to remove dust and contamination before applying any primer or coating material.

4.8.4.2 Peel adhesion method. Three inch by three inch by approximately 0.04 inch fiberglass panels shall be used for this test. After sanding and solvent cleaning as specified in 4.8.4.1, a two by three inch part of the panel shall be covered with Tedlar film approximately 0.002 inch thick held down at the exposed fiberglass surface by a 1/2 inch piece of pressure sensitive cellulose tape on both sides. The entire surface shall be coated as specified in 4.7.2. After 4 days, the Tedlar film and cellulose pressure sensitive tape shall be carefully removed. Two panels shall be cured for a total of 168 hours and two panels bake cured as specified in 4.7.2. The panels shall be immersed in water for 4 hours. Four parallel razor blades cut through to the fiberglass shall be scribed in the panel crossing the nonadhered and adhered coating sections and placed at 1/4 inch, 1-1/4 inch, 1-3/4 inch and 2-3/4 inch from the edge of the panel. One 1 inch free strip of coating shall be folded back over the adhered coatings and the uncoated panel and free strip of coating shall be placed in the jaws of a Scott Tensile Tester to provide 180° pull at a speed of 2 inches per minute.

4.8.5 Strength of free films. Two 8 inch by 10 inch sheets of unprocessed photographic glossy surface print or enlarged paper shall be coated on the emulsion side with the test coating as specified in 4.7.2. After 4 days cure, the sheet of paper shall be placed in luke warm water and the photographic paper and free film of coating separated. After blotting dry with paper towels, the films shall be cured and baked as specified in 4.7.2 and tested according to ASTM D882-67 Method A for the requirements of 3.7.4.

4.8.6 Low temperature flexibility. Two panels as specified in 4.7.1 shall be coated and room temperature cured as specified in 4.7.2. The coated panels shall then be conditioned for a minimum of 1 hour at -60°F and while at that temperature bent 180 degrees over a 1/4 inch mandrel in accordance with Method 6221 of Federal Test Method Std. No. 141. The panels shall be examined immediately after bending for evidence of failure, then conditioned at 70 ±5°F, and reexamined for conformance to 3.7.9.

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4.8.7 Water resistance. Two panels as specified in 4.7.1 shall be coated and room temperature cured as specified in 4.7.2. The panels shall be immersed in distilled water for 4 hours at 73 ±5°F. At the end of the immersion period, the specimens shall be removed from the water and examined for conformance to 3.7.10.

4.8.8 Accelerated weather resistance. Test panels prepared as directed in 4.7, shall be exposed for 500 hours in a 6000 watt xenon-arc weatherometer 1/ that is cycling between 102 minutes of light only and 18 minutes of light and waterspray. Coatings shall be examined to determine conformance to the requirements of 3.7.13. The following conditions shall apply when tested according to ASTM G 26, Type BH:

Black body temperature in cabinet: 140±5°F
Relative humidity in cabinet: 50±5°F
Intensity of Xenon-arc: 0.5-0.6 watts/sq. meter at 340 nm wavelength

1/ Atlas Electric Devices Company or equal

4.8.9 Electrical transmission.

4.8.9.1 Test panels. Test panels shall be approximately 24 inches by 34 inches by 0.165 ±0.01 inch thick. They shall be fabricated using 16 layers of No. 181E Fiberglass and polyester resin conforming to L-P-383. The fiberglass to polyester ratio shall be 62 to 38 parts by weight. The laminate shall be dense, of low void content and representative of high quality laminate fabrication. The dielectric constant of the laminate shall be 4.20 ±0.05. Coating shall be applied to the previously sanded and solvent wiped panel in accordance with 4.7.2 and 4.8.4.1. One panel shall be used as a control panel and shall be uncoated.

4.8.9.2 Sequence of testing. The sequence of testing the panels shall be as follows:

a. Coated panels shall be cured in accordance with 3.7.14.

b. The uncoated control panel and one coated panel shall be tested as specified below.

c. The same panels specified in (b) above shall be conditioned at a temperature of 100 ±5°F and a relative humidity of 90 ±5% for 7 days and then tested as specified below.

4.8.9.3 Transmission test equipment. Test equipment performance requirements for flat panel samples shall meet the transmission efficiency test of MIL-R-7705.

4.8.9.4 Test procedure. The test shall be conducted at a frequency of 9.375 (GHz) gigahertz. The test sample shall be clamped at the approximate midpoint between the horns at a 60° incident angle. The horns shall be adjusted so that polarization is horizontal. The incident angle shall be

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varied slightly on either side of 60° to maximize power transmission. Minimum and maximum power transmission shall be determined through a 3/8 inch lateral movement of the test panel and averaged. The average of the two values shall be considered the power transmission.

4.8.9.5 Test data and computation. The test data shall be converted to, and reported as, percent transmission, using the equation listed below (see 3.7.14):

$$\frac{\text{T2 sample}}{\text{T2 blank}} \times 100 = \text{percent transmission}$$

Where T2 sample = power transmission of coated panel

T2 blank = power transmission of uncoated panel

4.8.10 Hydrolytic stability. Two panels, as specified in 4.7.1, shall be coated and room temperature cured as specified in 4.7.2. The panels shall be stored at 200 ±5°F and 95 percent relative humidity for 4 days. A glass desiccator jar may be used for storage of the panels. The panels shall be placed, coated side up, on a tray in the desiccator. The desiccator shall contain a glycerine (22 percent by weight) in water solution in the bottom, which will produce a relative humidity (RH) of 95 percent at the test temperature. The desiccator shall then be closed and inserted into an air circulating oven maintained at 200 ±5°F. At the end of the exposure period inspect the panels visually and check for softening of the film by pushing a thumbnail or spatula over the film (see 3.7.15).

4.8.11 Surface resistivity (Type II only). Tests for surface resistivity shall be in accordance with Table VIII (see 3.7.16).

4.8.11.1 Equipment. The equipment used for the surface resistivity test shall be as specified in Figure 3; two 3/8 inch wire spring loaded electrodes having a pressure capacity of 3 to 5 pounds shall be as specified in Figure 3. Use a 500-volt megohmeter to measure accurately the range of 0 to 1000 megohms. The acrylic plastic panel shall conform to MIL-P-5425 and shall have a pattern of 12 holes, 1/2 inch in diameter, drilled in three rows of 4 holes spaced 6 inches apart, center to center. The first hole shall start at a point 3 inches from the side edge and 6 inches from the top edge. This template shall be used as a means of spacing the electrodes at equal distance for the resistivity readings.

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TABLE VIII. Surface Resistivity (Type II only).

Property	After drying & normal 7-day cure	After normal cure and aging 7 days at 95% RH & 100° ±2°F	After normal cure and accelerated weathering	After normal cure and aging 24 hours at 225° ±5°F
Surface resistivity (In megohms per square)	0.5 to 15	0.5 to 15	0.5 to 15	0.5 to 15

4.8.11.2 Test Panels. The panels shall be as specified in 4.8.9.1 except the size shall be 24 inches by 24 inches by 0.050 to 0.055 inch thick. The panels shall be coated and cured as specified in 4.7.2.

4.8.11.3 Measurement procedure. The template shall be placed over an individual panel which has been exposed to the conditions as specified in 4.7.2. The electrodes with their wire leads connected to the 500-volt megohmeter shall be placed to coincide with two holes of the template. All readings shall be taken with a uniform pressure of 3 to 5 pounds on the electrodes. Seventeen different readings shall be taken between holes of the template and averaged. After 1 hour, a repeat 17 readings shall be taken and averaged.

4.8.12 Pot life. The Type I and the Type II components shall be mixed as per manufacturer's instructions. The coatings shall be stored in a full closed container for 2 hours at room temperature. The working properties and surface appearance of a panel sprayed with the material shall conform to 3.7.3.

4.8.13 Applied film properties. The film shall be visually examined under 10x magnification for pinholes, cracks, bubbles or other irregularities (see 3.7.8).

4.8.14 Fuel resistance. Four coated panels shall be prepared and cured, as specified in 4.7.2. The panels shall be immersed in JP4 and JP5 fuel conforming to MIL-T-5624 for a period of 4 hours at room temperature. At the end of the period, the specimens shall be removed from the fluids and visually examined for conformance to paragraph 3.7.11.

4.8.15 Fluid resistance test. Six coated panels shall be prepared and cured, as specified in 4.7.2. The panels shall be immersed in each of the following for 4 hours: MIL-L-23699 lubricating oil, MIL-H-5606 hydraulic fluid and MIL-H-83282 hydraulic fluid. Four hours after removal, the films shall be visually examined for conformance to paragraph 3.7.12.

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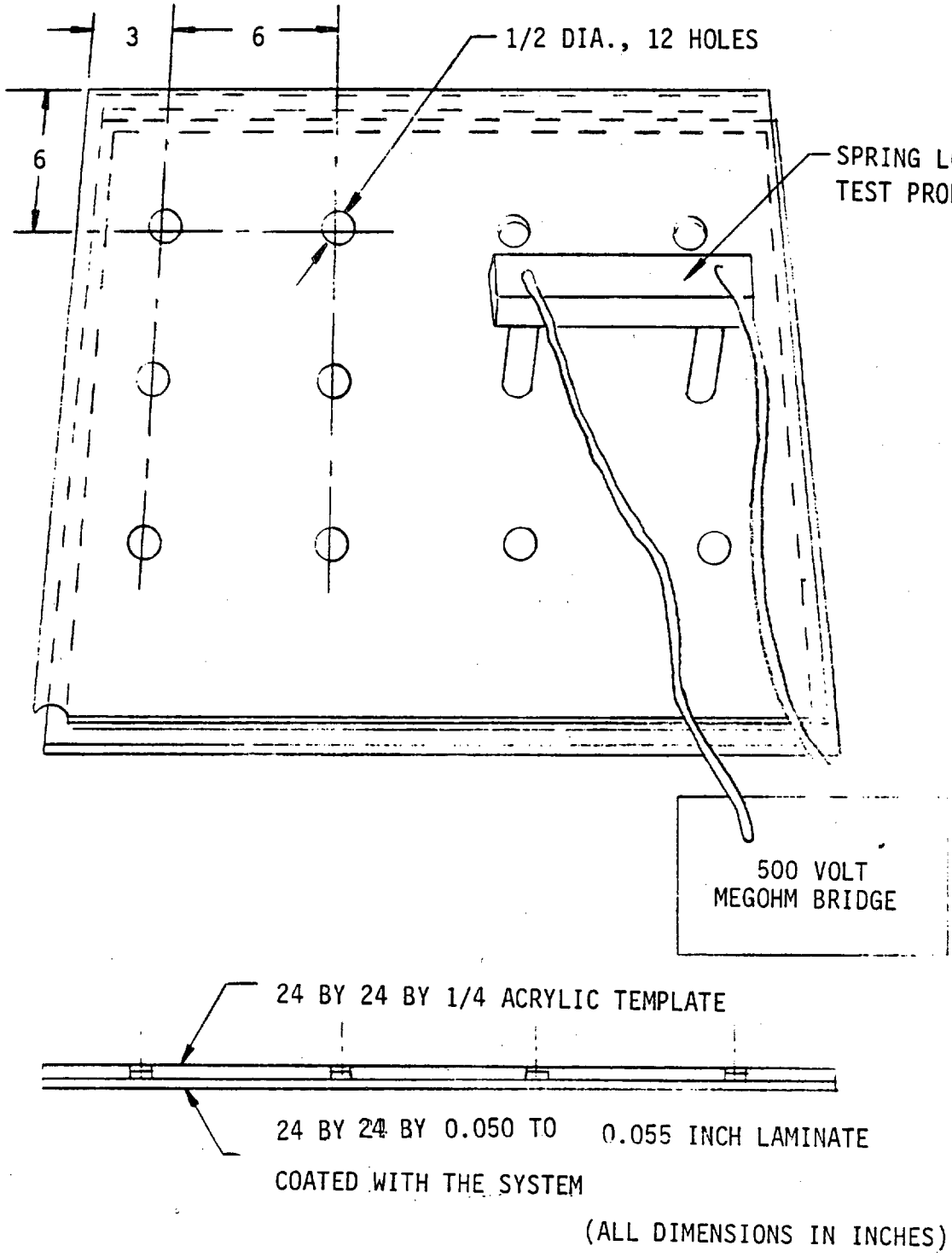


FIGURE 3. Apparatus used for surface resistivity.

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4.8 16 Strippability test. The Type I and Type II coatings shall be applied to flat, glass-fiber base laminates conforming to MIL-P-25421 and cured as specified in paragraph 4.7.2. Apply stripping compounds or equivalent 1/ and allow to stand approximately 15 minutes or until blistering and lifting occurs. Remove the softened coating using a plastic scrapper. Reapply stripping material to remove any residual primer or topcoat. Rub the test panel surface with a clean cloth saturated with methyl ethyl ketone (MEK) to clean the coating residue from the panel. The panel shall meet the requirements of paragraph 3.7.19.

1/ Gaco S-14 or equivalent Gaco SR-14 or equivalent	Gates Engineering Co. Wilmington, Delaware
B & B 5050L or equivalent	B & B Chemical Co., Inc. Hialeah, Florida
Pennwalt Stripper 780A or equivalent	Pennwalt Corp. Philadelphia, PA

4.8.17 Rain erosion resistance. Four round disks (1 inch in diameter by 3/16 inch thickness) cut from an epoxy-fiberglass laminate conforming to MIL-P-25421 shall be used for testing each color (gray, black or white) and type of coating. The coating shall be applied to the test panels as specified in paragraph 4.7.2 and cured as specified. Each coated disk shall be mounted in a horizontal whirling arm tester at the end of the rotating arm, so that the disk is perpendicular to the direction of travel. A suitable water ring, mounted above the rotating blade, shall be used to simulate a natural rainfall of one-half inch per hour and two millimeter droplet size. The test shall be run at 500 miles per hour at the sample's center (800 kilometers/hrs.). Determine the average time in which the 4 specimens of the same color and type have eroded through to the substrate (see 3.7.17).

4.8.18 Abrasion resistance test. The Type I and Type II coatings applied to flat, glass-fiber base laminates conforming to MIL-P-25421 and cured as specified in paragraph 4.7.2 shall be subjected to a gas carried stream of crushed glass beads of approximately 50 micron size (0.002 in) (S.S White Co. No. 10 or equal) through an orifice nozzle, 0.018 in (0.045 cm) in diameter which is fixed perpendicular to and at a distance of 0.625 in (1.58cm) from the test specimen using a standard measuring gauge. Dry, filtered, compressed air controlled at 60 psi (413.7KPa) pressure shall be used as the carrying gas. The test shall be conducted at least one inch (2.54 cm) from the edge of the test panel and not less than one half inch (1.27 cm) from an adjacent test site. To obtain an accurate film thickness relative to each test site, a strip of coating shall be removed carefully and measured using a film thickness guage (see 3.7.18).

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4.8.18.1 Abrasion resistance values. Abrasion resistance values shall be reported as the time, in seconds, required to penetrate a measured film as well as a numerical index value based on the following formula:

$$\text{ABRASION INDEX} = \frac{t_1}{S_1} \times \frac{S_2}{t_2}, \text{ where}$$

t_1 = time in seconds to penetrate the film under test
t_2 = time in seconds to penetrate the standard polyurethane film
S_1 = thickness of the film under test in mils (X 0.0025 = cm)
S_2 = thickness of the standard polyurethane film in mils (X0.0025 = cm)

The standard polyurethane film used as a control and check on the S. S. White apparatus is 12 mil (0.030 cm) 3M Scotch Tape, Type Y 9265 or equal. The coating shall meet the minimum requirements specified in Table V.

5. PACKAGING

5.1 Packaging and packing. The packaging, packing and marking of the coating components and kits shall be in accordance with PPP-P-1892.

5.1.1 Unit packaging. Unless otherwise specified, the coatings shall be packaged in quart size cans.

5.1.1.1 Rain erosion coating kit. The rain erosion coating components shall be packaged in separate containers. The rain erosion kit shall consist of two 1-quart cans of the same colored component A (resin portion) and one 1-quart can of component B (curing agent). The three containers shall be packed in a fiberboard box and identified as the rain erosion kit.

5.1.1.2 Antistatic coating kit. The antistatic coating components shall be packaged in separate containers. The antistatic kit shall consist of two 1-quart cans component A (resin portion) and one 1-quart can of component B (curing agent). The three containers shall be packed in a fiberboard box and identified as the antistatic kit.

5.2 Mixing and safety instructions.

5.2.1 Topcoat. In addition to the labels specified in PPP-P-1892, each topcoat component can shall contain the following mixing and spraying instructions and precautions.

"Precautions are already on label as required by OSHA.

1. Open Pigmented Component carefully. Discard if can is bulged.

2. Clean spray equipment and hoses thoroughly, before and immediately after use, with methyl ethyl ketone.

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3. Mix only that amount which can be used or within 2 hours.

4. Wear protective clothing, creams, and goggles while spraying.

5. A fresh air hood is recommended for spraying operations. If a fresh air hood is not available, protective masks containing an organic vapor cartridge respirator or airline respirator should be used.

6. Do not contaminate Components A or B with alcohol or water.

7. Keep containers closed when not in use.

8. Wash hands before eating or smoking."

5.2.1.1 Label on lid of pigmented component. The container lid of the pigmented component shall bear a statement in red: "USE CAUTION -- OPEN CAREFULLY".

6. NOTES

6.1 Intended use. The material covered by this specification is intended as a rain-erosion coating for aircraft radomes and leading edges. The coating can also be utilized as a flexible sealant to protect fastener patterns on the exterior surfaces of aircraft.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type desired (see 1.2).

c. Components, specofy when different (see 3.3).

d. Color number and name. Requisitions, contracts, and orders should specify the quantity and size of container or containers in which the polyurethane coating is to be furnished.

e. The unit of purchase to be made.

f. Level of packaging and packing (see Section 5).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirment; and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the U. S. Naval Air Development Center, Attn: Code 60622, Warminster, PA 18974; and information pertaining to qualification of products may be obtained from that activity.

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6.4 Moisture. The polyurethane components should be kept dry. The presence of moisture degrades the quality of the paint. Packaging of the materials should be done in a dry atmosphere. Solvents for the resins should be examined for evidence of contamination before they are incorporated even though they are of urethane grade. Urethane grade solvents or thinners may become contaminated with water in tank cars or storage tanks. The purchase of urethane grade solvents or thinners is no guarantee that excessive moisture is not present. It is therefore recommended that all users check for moisture contamination. The following suggested method may be used to determine the presence of water: add one drop of aluminum secondary butoxide to 100 ml of the solvent in a stoppered flask and shake. An appreciable amount of turbidity indicates the presence of water. The pigments used in the topcoats must be absolutely dry before being incorporated into the resin solution. If not, the package stability will suffer.

6.5 Ready-to-recoat. A film shall be considered ready for recoating when a subsequent coat can be applied without the development of any film irregularities, such as lifting or loss of adhesion of the first coat, and the drying time for the second coat does not exceed the maximum specified for the first coat.

Custodians:
Army - ME
Navy - AS
Air Force - 11

Review activities:
Army - AV, ME

User activities:
Navy - MC

Preparing activity
Navy - AS
(Project No. 8010-0938)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-85322	Coating, Elastomeric, Polyurethane, Rain Erosion Resistant, For Exterior Aircraft Use

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation.

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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